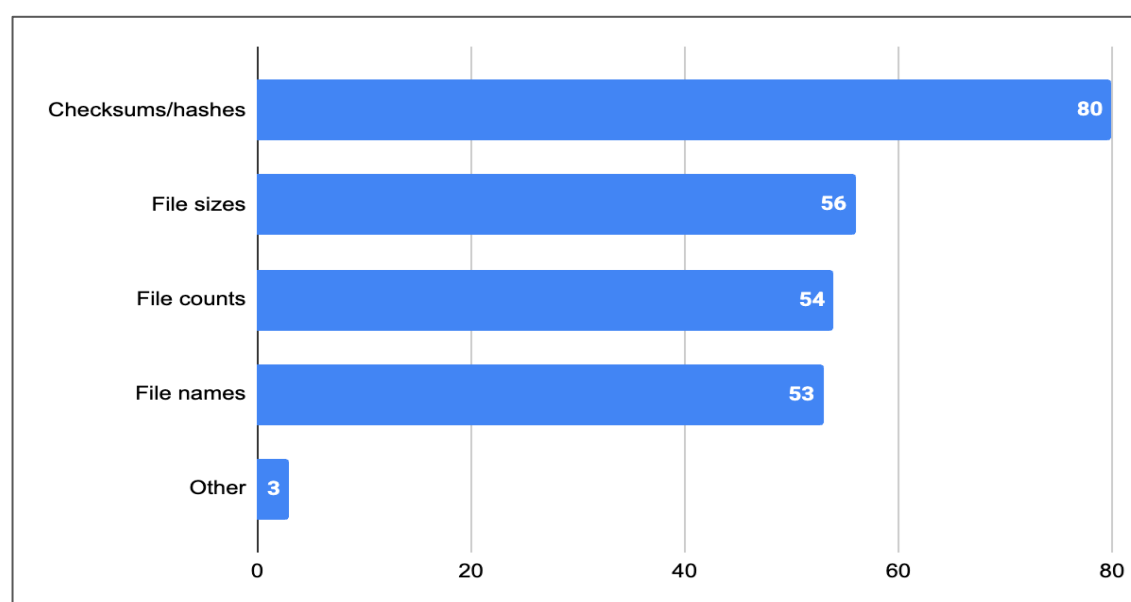


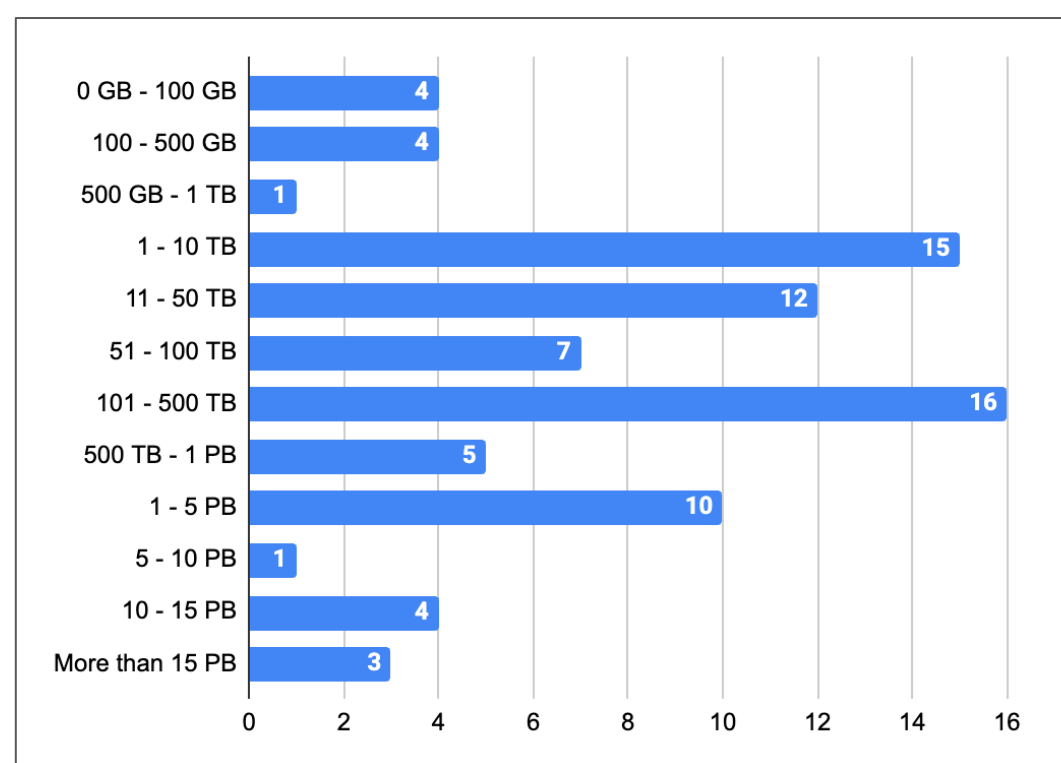
Survey Overview

- This longitudinal study builds on previous surveys conducted in 2017 and 2021
- Survey administered from July 15th to September 15th 2025
- Open to both members (39%) and non-members (61%)
- A total of 44 questions, of mixed formats
- A total of 152 responses were recorded, of which 89 completed in the report
- 76% of respondents were located in the United States, with 8 other countries represented

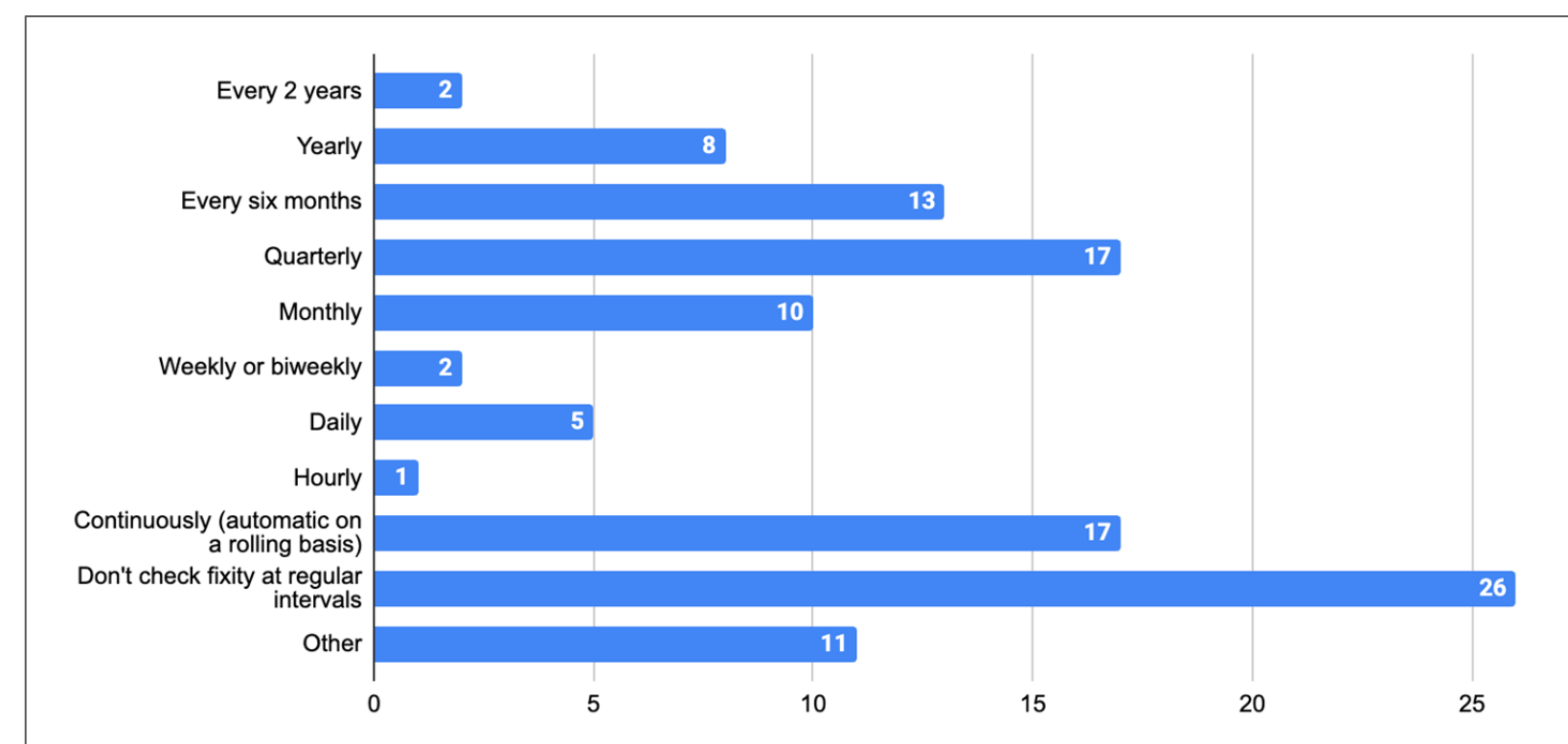
94% of respondents use fixity at some point; MD5 was the most commonly used



Total Volume of Content



Frequency of Fixity Checking Activities



Reasons / Methods of Fixity Practices

"Generating fixity via bags is a standard part of our accessioning (initial work done within 4 months of receipt). It gives us the longest authenticity trail and makes it easier to monitor that content is unaltered while it is in storage awaiting further processing."

"The preserve the digital integrity of the files. Files are transferred a few times before they are stored in our digital repository, and we need to validate that the files have not changed, degraded, been deleted, etc."

This poster shares the results from the 2025 Fixity Survey and illuminates the value of community-wide data gathering and knowledge sharing.

The survey results demonstrate a large variation in fixity practices across the community including, when fixity information is captured or verified, the frequency of checks, where fixity information is recorded, and which checksum algorithms are in use.

Fixity and the Cloud

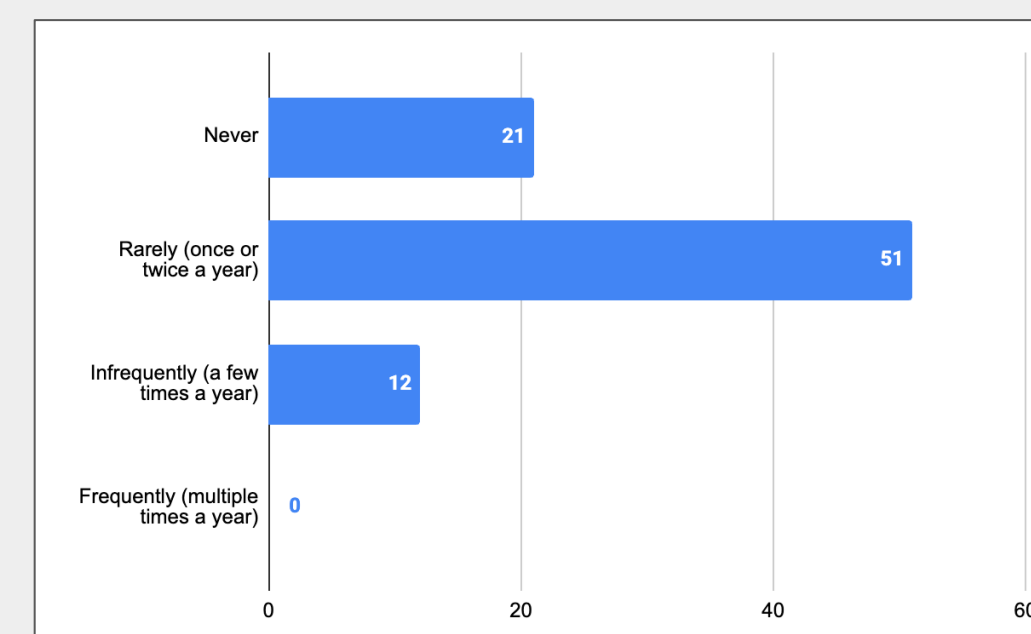
63% use cloud, 55% can run fixity on the cloud, 58% receive information from cloud service, 72% use this info, but most don't save the information

Follow-up interviews - Case Studies - Provide insight into fixity practices at:

- Special Collections within a library of a liberal arts college
- State Archives, responsible for government records and special collections
- Federal government office in compliance with ISO16363

Fixity Failures: Not common; multiple methods for reviewing and addressing

The most common errors were corrupted bit streams or interrupted network transfers.



"When our IT Department transferred the Digital Archives storage to new servers, we ran Integrity Checks and found a couple of hundred files from a specific document store failed the check. We were able to educate the IT staff about checksums and showed them that even though they robocopied the files - [they] had changed. The IT staff replaced the document store and the check was successful."

Fixity Checking Frequency Factors

Extremely Important	Number of Responses
Time it takes to run fixity check	34 out of 79, or 43%
Available staff	29 out of 79, or 36.7%

Not Important	Number of Responses
Terms of access by service providers	35 out of 78, or 44.9%
Concern of media failure due to increased use of storage media	34 out of 80, or 42.5%
Regular checks done at the block level via a system	33 out of 78, or 42.3%

Key Takeaways

- The primary reason the community uses fixity information is to determine whether data has been altered over time.
- While there is consensus that using fixity information represents good practice, the survey results demonstrate substantial variation in fixity practices across the community. There are a variety of practices reported, including when fixity information is captured or verified, the frequency of checks, where fixity information is recorded, and what checksum algorithms are in use.
- Addressing fixity errors involved confirming whether the result was an error, troubleshooting errors, and replacing the incorrect file with a known-good copy.

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Link to Full Report:

For background on NDSA, visit:

<https://ndsa.org/>

